

NOTES: Turning Hamiltonians into Images

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1 Introduction

Given a Hamiltonian, we want to find a unitary transformation which makes the Hamiltonian as “smooth” and “image-like” as possible. Before proposing a method to achieve this, we need to define what we mean by “smooth” and “image-like”. Once we have a measure of smoothness or roughness, our goal will be to find a unitary transformation which maximizes or minimizes this measure respectively.

We will first begin with symmetric matrices with real entries and try to find orthogonal transformations which will smoothen the matrix. Then we will move on to hermitian matrices with complex entries and their unitary transformations.

2 Defining smoothness

Let H be an $n \times n$ symmetric matrix with real entries. We define the roughness of the matrix $R(H)$ by summing the squares of differences between adjacent entries in the matrix.

$$R(H) = \sum_{i=1}^{n-1} \sum_{j=1}^n (H_{i,j} - H_{i+1,j})^2 + \sum_{i=1}^n \sum_{j=1}^{n-1} (H_{i,j} - H_{i,j+1})^2 \quad (1)$$

where $H_{i,j}$ is the (i,j) th entry of the matrix H . The roughness is a measure of how much the entries of the matrix vary from their neighbors. The roughness is zero if all the entries of the matrix are equal, and it increases as the entries of the matrix become more different from each other.

This definition can be translated to a more concise form by defining a matrix R as follows:

$$R_{i,j} = \begin{cases} -1 & \text{if } i = j < n \\ 1 & \text{if } i = j + 1 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where n is the size of the matrix. The roughness of the matrix can then be expressed as:

$$R(H) = \|RH\|^2 + \|HR^T\|^2 \quad (3)$$

where $\|\cdot\|$ is the Frobenius norm. For symmetric matrices, $R(H)$ further simplifies to:

$$R(H) = 2\|RH\|^2 \quad (4)$$

3 Problem Simplification

Spectral Theorem: Let A be an $n \times n$ matrix with real entries. Then, the following statements are equivalent:

1. A is symmetric
2. A is orthogonally diagonalizable
3. A has n orthogonal eigenvectors

Using the spectral theorem, we can further simplify the problem. Given any symmetric matrix H , spectral theorem guarantees the existence of an orthogonal transformation that will diagonalize the matrix. Now our goal is to smoothen this diagonal matrix. So the problem simplifies as follows:

“Given a diagonal matrix D with real entries, find an orthogonal transformation U such that $R(UDU^T)$ is minimized.”

4 Exact Solution for 2x2 Hamiltonians

Let D be a 2×2 diagonal matrix with real entries. We can write the matrix as:

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad (5)$$

where λ_1 and λ_2 are the eigenvalues of the matrix.

We can write any arbitrary orthogonal transformation in $\mathbb{R}^2$ as:

$$U = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \quad (6)$$

where θ is the angle of rotation. The transformed matrix can be written as:

$$\tilde{D} = U^T D U = \frac{1}{2} \begin{pmatrix} (\lambda_1 + \lambda_2) + (\lambda_1 - \lambda_2) \cos(2\theta) & (\lambda_2 - \lambda_1) \sin(2\theta) \\ (\lambda_2 - \lambda_1) \sin(2\theta) & (\lambda_1 + \lambda_2) - (\lambda_1 - \lambda_2) \cos(2\theta) \end{pmatrix} \quad (7)$$

We can now compute the roughness of the transformed matrix:

$$R(\tilde{D}) = 2(\lambda_1^2 + \lambda_2^2) + 2(\lambda_1^2 - \lambda_2^2) \sin(2\theta) \quad (8)$$

By analyzing the first and second derivatives of the roughness function, we can find the minimum value of the roughness is attained at $\theta = \frac{\pi}{4}$ if $|\lambda_1| < |\lambda_2|$ and $\theta = \frac{3\pi}{4}$ if $|\lambda_1| > |\lambda_2|$. The minimum value of the roughness is given by:

$$R_{min} = 4 \min\{\lambda_1^2, \lambda_2^2\} \quad (9)$$

Note that if $|\lambda_1| = |\lambda_2|$, then the roughness function is constant and hence cannot be minimized by any orthogonal transformation.

5 Algorithmic Approach in Higher Dimensions

Let us define the following notations:

- $M(n, \mathbb{R})$ = set of all $n \times n$ matrices with real entries
- $Sym(n, \mathbb{R})$ = set of all $n \times n$ symmetric matrices with real entries
- $O(n)$ = set of all $n \times n$ orthogonal matrices

Let $H \in Sym(n, \mathbb{R})$ be the Hamiltonian we want to smoothen.

Define a function $f : M(n, \mathbb{R}) \rightarrow \mathbb{R}$ as follows:

$$f(U) = R(U^T H U) = 2\|R U^T H U\|^2 \quad (10)$$

$$= 2Tr[(R U^T H U)^T R U^T H U] = 2Tr[U^T H U R^T R U^T H U] = 2Tr[H U R^T R U^T H] \quad (11)$$

We want to minimize the function $f(U)$ subject to the constraint that $U \in O(n)$. This is a constrained optimization problem on the smooth manifold of orthogonal matrices. Hence we can use the method of Riemannian gradient descent^[1](RGD) to find the minimum of the function. The algorithm is as follows:

1. Initialize $U_0 \in O(n)$
2. At any k^{th} step, compute the Euclidean gradient of the function $f(U)$ at U_k :

$$\nabla f(U_k) = 2H^2 U_k R^T R \quad (12)$$

3. Compute the Riemannian gradient of the function $f(U)$ at U_k by projecting the Euclidean gradient onto the tangent space of $O(n)$ at U_k :

$$grad f(U_k) = \frac{1}{2} (\nabla f(U_k) - U_k \nabla f(U_k) U_k^T) \quad (13)$$

4. **Retraction:** Move along a smooth curve on the manifold $O(n)$ along the direction of the Riemannian gradient to reach the next point U_{k+1} . Retraction is a map from the tangent space of the manifold to the manifold that does exactly the same. Note that the retraction is not unique and can be chosen based on the problem at hand. In our case, we can use the QR decomposition of $U_k - \eta grad f(U_k)$ to get the next point on the manifold, where η is the learning rate. The retraction can be expressed as:

$$U_{k+1} = Q, \text{ where } U_k - \eta grad f(U_k) = QR \text{ is the QR decomposition} \quad (14)$$

5. Repeat steps 2-4 until convergence.

6 Results

6.1 Smoothing Hamiltonians

Let us use the algorithm to smoothen a Hamiltonian. We will use a 6×6 Hamiltonian which has a single non-zero entry and then smoothen it.

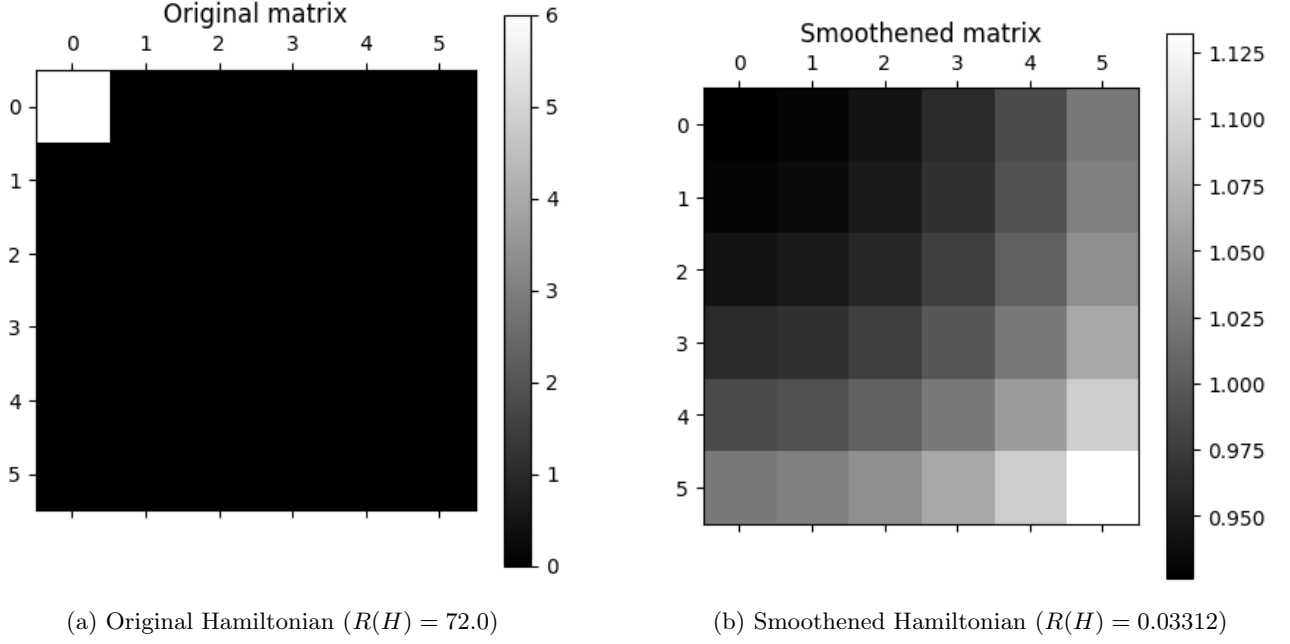


Figure 1: Array plots of the original and smoothed Hamiltonians.

Now let us use the algorithm to smoothen a randomly generated diagonal matrix. We will use a 7×7 Hamiltonian with random entries on its diagonal.

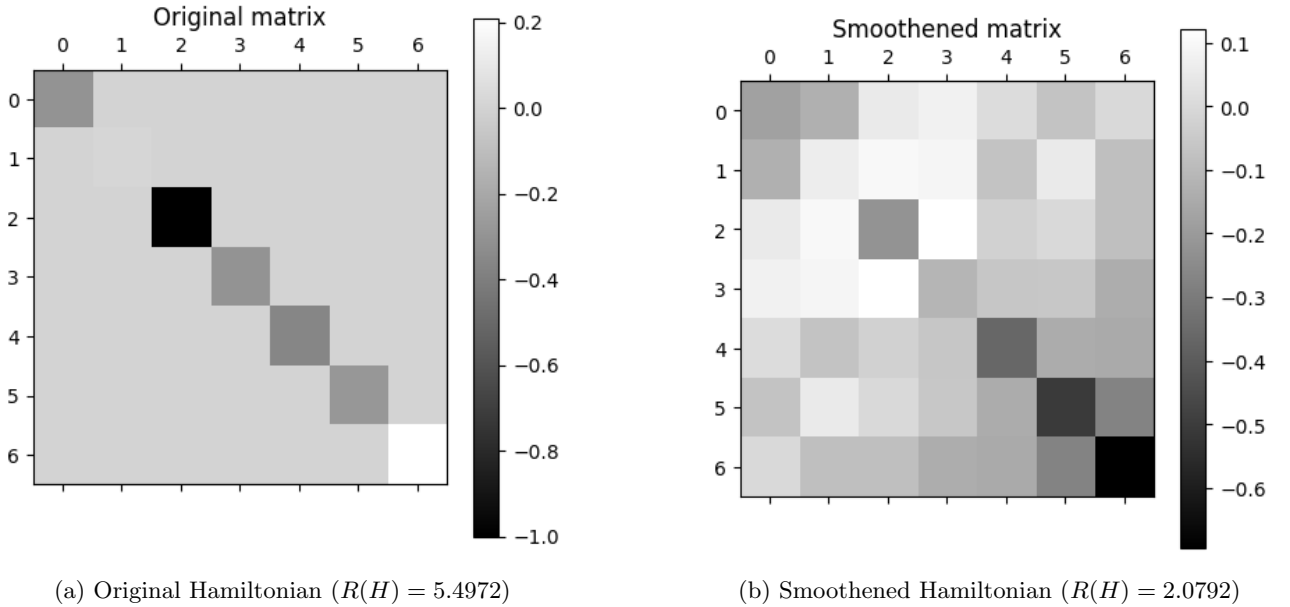


Figure 2: Array plots of the original and smoothed Hamiltonians.

6.2 Comparing with exact solution : 2×2 Hamiltonians

Let us use the algorithm to smoothen a 2×2 Hamiltonian and compare the results with the exact solution. Let H be the Hamiltonian given by:

$$H = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \quad (15)$$

Analytically, the smoothest form of the Hamiltonian is given by:

$$H_{smooth} = \begin{pmatrix} 2 & \sqrt{5} \\ \sqrt{5} & 2 \end{pmatrix} \approx \begin{pmatrix} 2 & 2.236 \\ 2.236 & 2 \end{pmatrix} \quad (16)$$

Computing the smoothest form of the Hamiltonian using the algorithm, we get:

$$\begin{bmatrix} [1.99412755 & 2.23606027] \\ [2.23606027 & 2.00587245] \end{bmatrix}$$

The algorithm reduced the roughness of the Hamiltonian from $R(H) = 4$ to $R(H_{smooth}) = 0.2230357$. We can see that the algorithm covered to the exact solution in a few iterations.

7 Challenges

- For arbitrary Hamiltonians of dimension greater than 7, the algorithm takes a long time to converge.
- We have fixed the learning rate to a constant value. This may not be optimal for all Hamiltonians.
- We used the QR decomposition to compute the retraction. This may not be optimal for all Hamiltonians. We can use other retraction methods such as polar retraction.
- RGD is a first order method, we might require a second order method for faster convergence in larger Hamiltonians.

8 Closed form solution using Lagrange Multipliers

Our goal is to minimize the function $f(U)$ subject to the constraint that $U \in O(n)$. We can use Lagrange multipliers for this. Let us define the Lagrangian as:

$$\mathcal{L}(U, \Lambda) = f(U) + Tr[\Lambda(U^T U - I)] = 2Tr[HUR^T RU^T H] + Tr[\Lambda(U^T U - I)] \quad (17)$$

$$= 2Tr[U^T AUB] + Tr[\Lambda(U^T U - I)] \quad (18)$$

where $A = H^2$ and $B = R^T R$.

Taking the derivative of the Lagrangian with respect to U and setting it to zero, we get:

$$\frac{\partial \mathcal{L}}{\partial U} = 2AUB + U(\Lambda + \Lambda^T) = 0 \quad (19)$$

Taking the derivative of the Lagrangian with respect to Λ and setting it to zero, we get the orthogonality constraint:

$$\frac{\partial \mathcal{L}}{\partial \Lambda} = U^T U - I = 0 \quad (20)$$

We can solve the above equations to get the optimal U and Λ .

$$2AUB = -U(\Lambda + \Lambda^T) \quad (21)$$

$$\implies 2U^T AUB = -(\Lambda + \Lambda^T) \quad (22)$$

$$\implies 2BU^T AU = -(\Lambda^T + \Lambda) \quad (23)$$

$$\implies [U^T AU, B] = 0 \quad (24)$$

Therefore, $U^T AU$ and B commute and hence can be simultaneously diagonalized. Note that B is a symmetric matrix and hence is orthogonally diagonalizable(Spectral theorem). So $\exists P \in O(n)$ such that $B = P^T D_B P$ and $U^T AU = P^T D_A P$ where D_A and D_B are diagonal matrices.

$$U^T AU = P^T D_A P \quad (25)$$

$$\implies A = UP^T D_A P U^T \quad (26)$$

Hence, PU^T is the orthogonal transformation that diagonalizes the matrix A . And, P is the orthogonal transformation that diagonalizes the matrix B . Let E_A and E_B be orthogonal matrices constructed from the eigenvectors of A and B respectively.

$$A = E_A^T D_A E_A, B = E_B^T D_B E_B \quad (27)$$

$$\implies PU^T = E_A, P = E_B \quad (28)$$

$$\implies U = E_A^T E_B \quad (29)$$

Hence, the optimal orthogonal transformation that minimizes the roughness of the Hamiltonian is given by:

$$U = E_A^T E_B \quad (30)$$

But the eigenvectors of A and B can be ordered in any way. So, let us find the optimal ordering of the eigenvectors of A and B such that the roughness is minimized.

Let us compute $f(U)$ -

$$f(U) = 2Tr[U^T A U B] \quad (31)$$

$$= 2Tr[U^T E_A^T D_A E_A U E_B^T D_B E_B] \quad (32)$$

$$= 2Tr[U^T U P^T D_A P U^T U P^T D_B P] \quad (33)$$

$$= 2Tr[P^T D_A P P^T D_B P] \quad (34)$$

$$= 2Tr[D_A D_B] = 2 \sum_{i=1}^n \lambda_i^A \lambda_i^B \quad (35)$$

Hence, the optimal ordering of the eigenvectors of A and B is such that the the sum of the products of the eigenvalues is minimized. By rearrangement inequality, this is achieved if eigenvalues of A increases and eigenvalues of B decreases, as we go down the diagonal in D_A and D_B respectively.

This gives us a closed form solution for the optimal orthogonal transformation that minimizes the roughness of the Hamiltonian.

So we can conclude that the optimal orthogonal transformation that minimizes the roughness of the Hamiltonian is given by:

$$U = E_A^T E_B \quad (36)$$

where E_A and E_B are the orthogonal matrices constructed from the eigenvectors of A and B respectively. The optimal ordering of the eigenvectors of A and B is such that the sum of the products of the eigenvalues is minimized. By rearrangement inequality, this is achieved if eigenvalues of A increases and eigenvalues of B decreases, as we go down the diagonal in D_A and D_B respectively.

9 Numerical Results

We can use the closed form solution to smoothen Hamiltonians of arbitrary dimensions. The following results were obtained using the closed form solution.

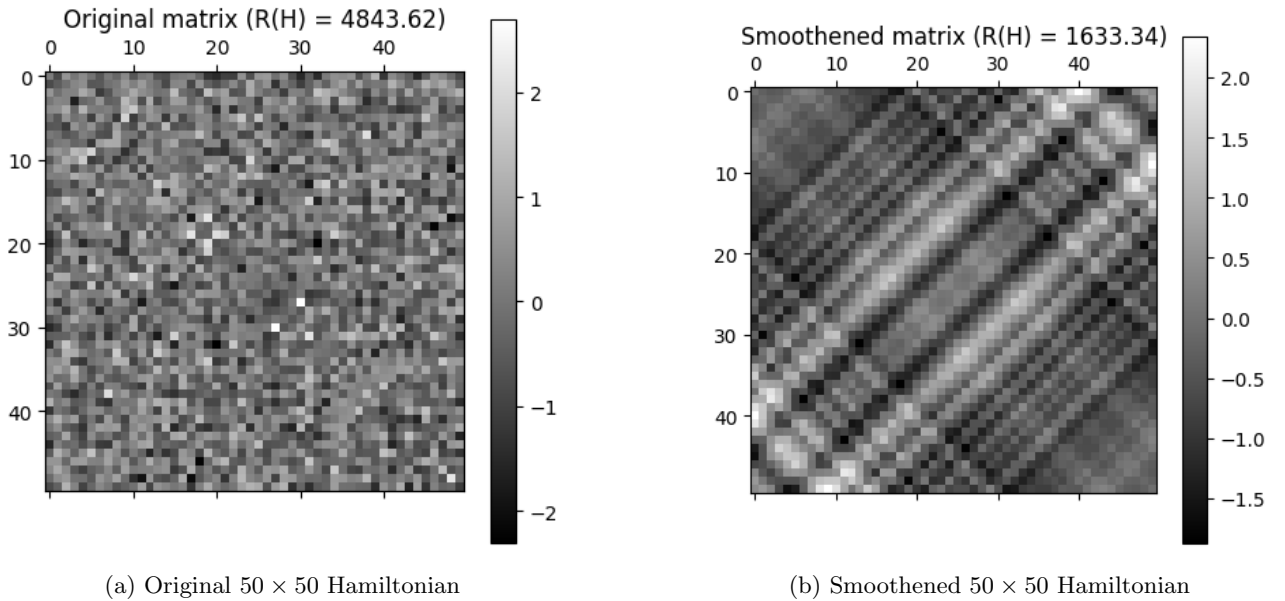
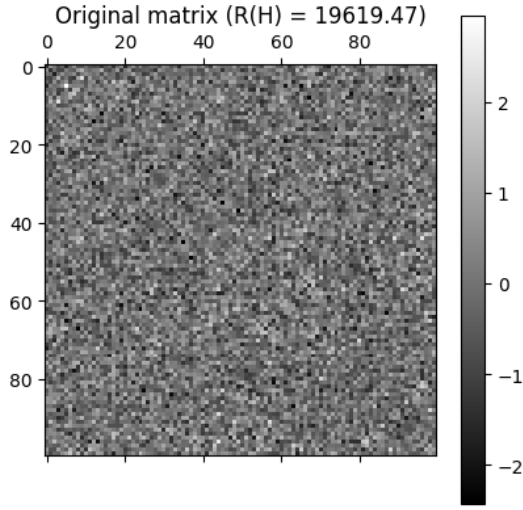
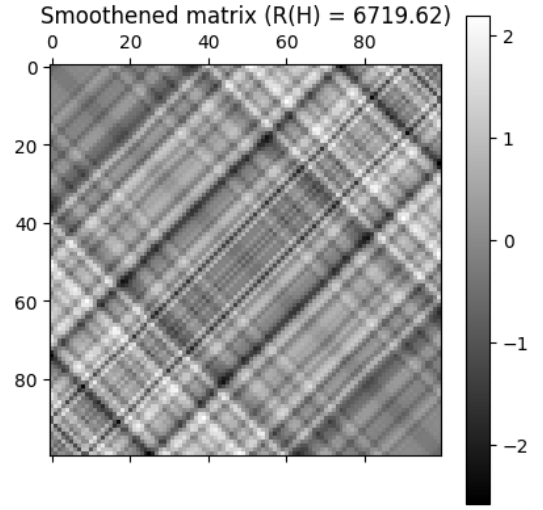


Figure 3: Array plots of the original and smoothened Hamiltonians.

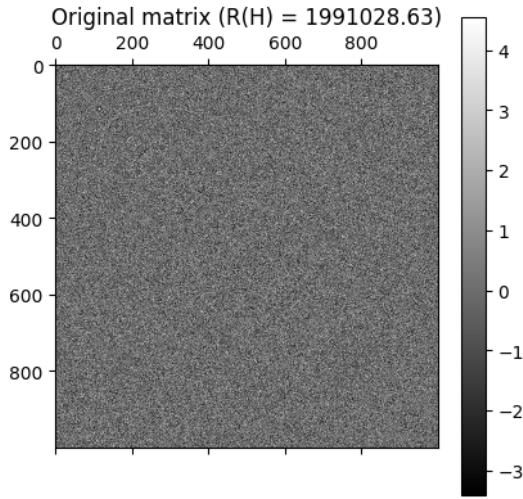


(a) Original 100×100 Hamiltonian

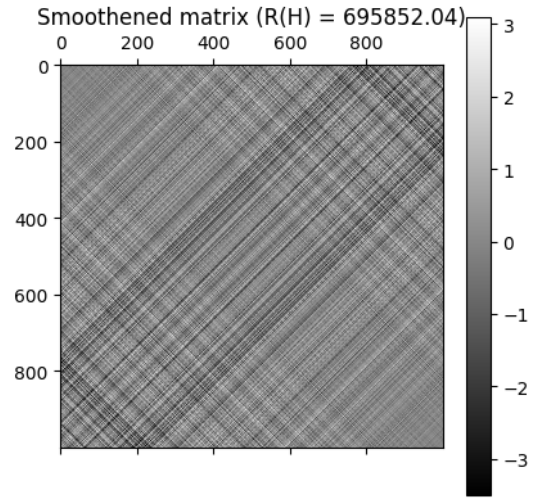


(b) Smoothed 100×100 Hamiltonian

Figure 4: Array plots of the original and smoothed Hamiltonians.



(a) Original 1000×1000 Hamiltonian



(b) Smoothed 1000×1000 Hamiltonian

Figure 5: Array plots of the original and smoothed Hamiltonians.

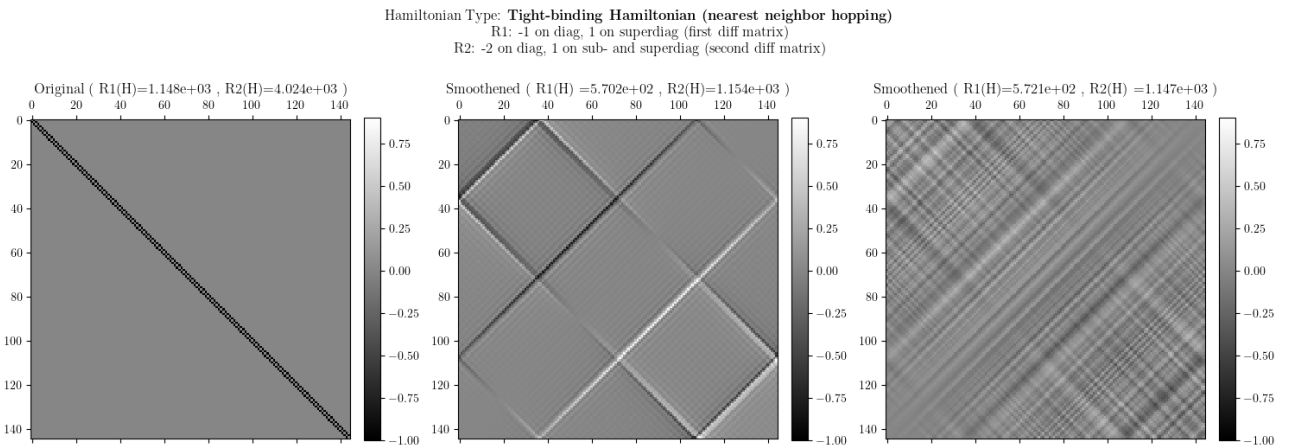


Figure 6: 145×145 Tight Binding Hamiltonian with nearest neighbor interactions (Hopping parameter : $t = -1$).

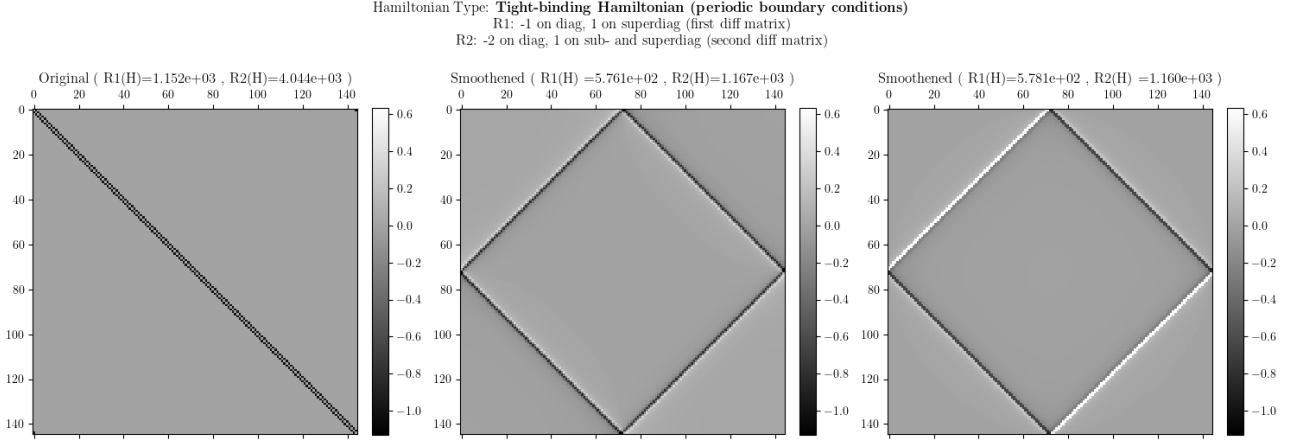


Figure 7: 145×145 Tight Binding Hamiltonian with nearest neighbor interactions and periodic boundary conditions (Hopping parameter : $t = -1$).

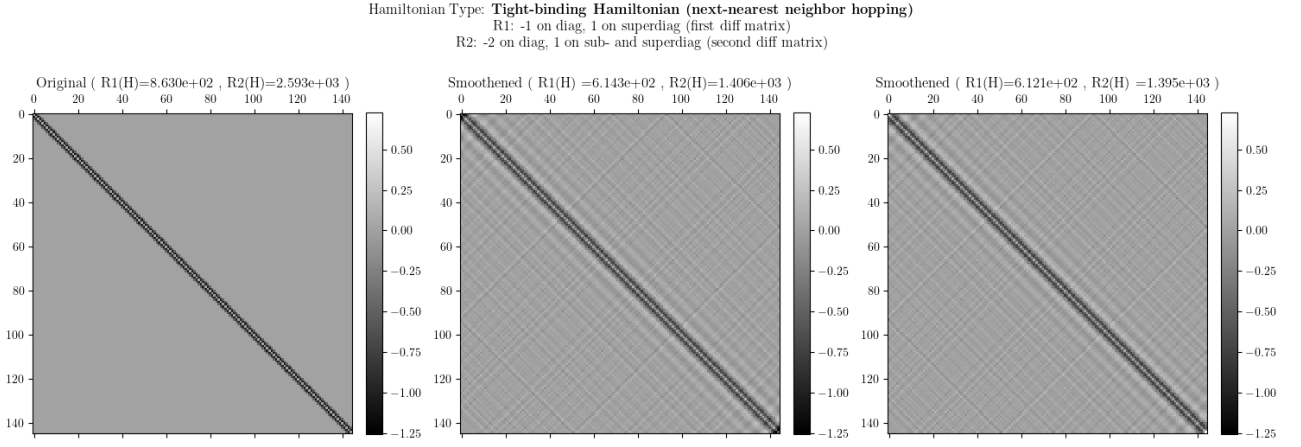


Figure 8: 145×145 Tight Binding Hamiltonian with nearest and next nearest neighbor interactions (Hopping parameter: $t_{nn} = -1, t_{nnn} = -1/2$).

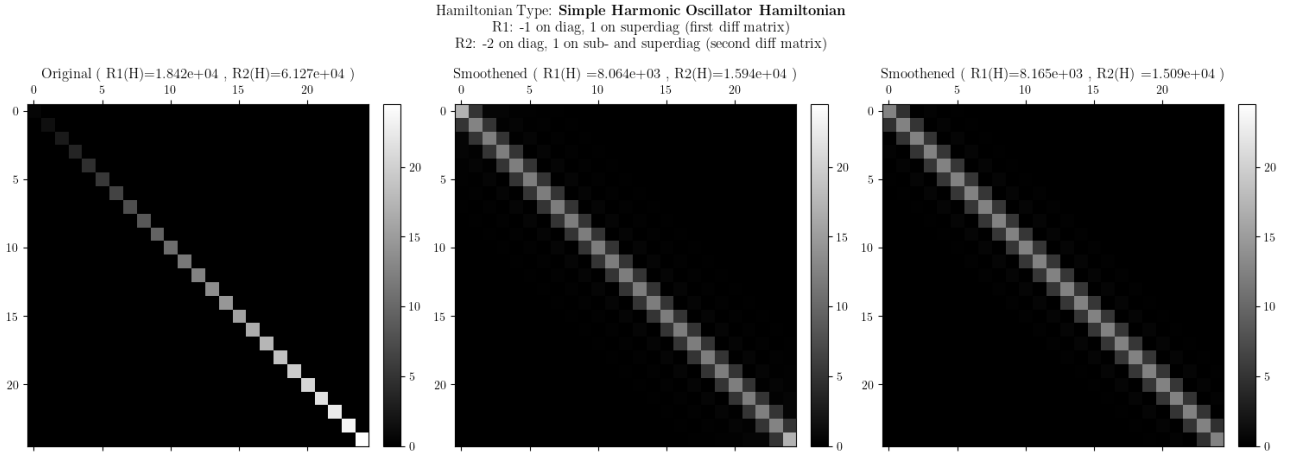


Figure 9: 25×25 Simple Harmonic Oscillator Hamiltonian with $m = 1, \omega = 1$ and $\hbar = 1$.

10 Uniqueness of the solution

In this section, we will explore the uniqueness of the smoothest form of the Hamiltonian.

10.1 Uniqueness of the smoothest representation

We had illustrated that for the optimal orthogonal transformation U ,

$$f(U) = 2 \sum_{i=1}^n \lambda_i^A \lambda_i^B \quad (37)$$

where, $\lambda_1^A \geq \lambda_2^A \geq \dots \geq \lambda_n^A$ and $\lambda_1^B \leq \lambda_2^B \leq \dots \leq \lambda_n^B$ are the eigenvalues of A and B respectively.

Definition: A matrix A is said to be “absolutely degenerate” if there exists a real number $\lambda \neq 0$ such that λ and $-\lambda$ are both eigenvalues of A .

Lemma: If H is a symmetric $n \times n$ matrix with real entries and H is not absolutely degenerate, then any eigenvector of $A = H^2$ is an eigenvector of H .

Proof

Let v be an eigenvector of H^2 with eigenvalue λ^2 . Since H is symmetric, there exists an orthonormal basis $\{q_1, q_2, \dots, q_n\}$ of eigenvectors of H such that $Hq_i = \mu_i q_i$ where μ_i are the eigenvalues of H . Then we can write:

$$v = \sum_{i=1}^n c_i q_i \quad (38)$$

$$\implies H^2 v = H^2 \left(\sum_{i=1}^n c_i q_i \right) = \sum_{i=1}^n c_i \mu_i^2 q_i = \lambda^2 v = \sum_{i=1}^n c_i \lambda^2 q_i \quad (39)$$

$$\implies \sum_{i=1}^n c_i (\mu_i^2 - \lambda^2) q_i = 0 \quad (40)$$

Since the eigenvectors of H are orthogonal, the above equation implies that for each i , $c_i (\mu_i^2 - \lambda^2) = 0$. This means that for $c_i \neq 0$, we must have $\mu_i = \pm \lambda$. Since H is not absolutely degenerate, we can assume without loss of generality that $\mu_i = \lambda$ whenever $c_i \neq 0$. Hence, v is a linear combination of eigenvectors of H with eigenvalue λ . This means that v is an eigenvector of H with eigenvalue λ .

Theorem: The smoothest form of a Hamiltonian H is unique if and only if the matrix H is not absolutely degenerate.

Proof

Case 1: H is not absolutely degenerate.

Let H be a Hamiltonian which is not absolutely degenerate. By the previous lemma, any eigenvector of $A = H^2$ is an eigenvector of H . Hence, any orthogonal transformation that diagonalizes A will also diagonalize H . That is, $E_A H E_A^T = D_H$ where D_H is the diagonal matrix with eigenvalues of H arranged in decreasing order of their absolute values. Since H is not absolutely degenerate, there is a unique way of ordering the eigenvalues of H in the decreasing order of their absolute values (Even if some eigenvalues are equal, exchanging them won't change D_H , it will only change the transformation E_A). Hence, D_H is unique and hence the smoothest images $E_B^T D_H E_B$ is also unique.

Case 2: H is absolutely degenerate.

Let H be a Hamiltonian which is absolutely degenerate. Then, there exists a real number λ such that λ and $-\lambda$ are both eigenvalues of H . Now, there isn't a unique way of ordering the eigenvalues of H in the decreasing order of their absolute values. We can exchange the eigenvalues λ and $-\lambda$ and still meet the optimal ordering condition. But this will produce two different D_H matrices, and hence two different smoothest images $E_B^T D_H E_B$. Hence, the smoothest form of the Hamiltonian is not unique.

10.2 Example: Absolutely degenerate Hamiltonian

Figure(10) shows different smoothest representations of an absolutely degenerate Hamiltonian. The first four images are obtained by orthogonal transformations E_A that diagonalizes H and H^2 simultaneously, then we exchange λ and $-\lambda$ in the diagonal representation of H to get the four different smoothest representations. The last two images are obtained by transformations E_A that diagonalizes H^2 but not H . This is done by taking linear combinations of the eigenvectors of H with eigenvalues λ and $-\lambda$ and then using them to construct the orthogonal transformation E_A .

We can see that the first four images have anti-diagonal symmetry, but the last two images do not have this symmetry. We will later see that this symmetry is a salient feature of the smoothest form when E_A diagonalizes H .

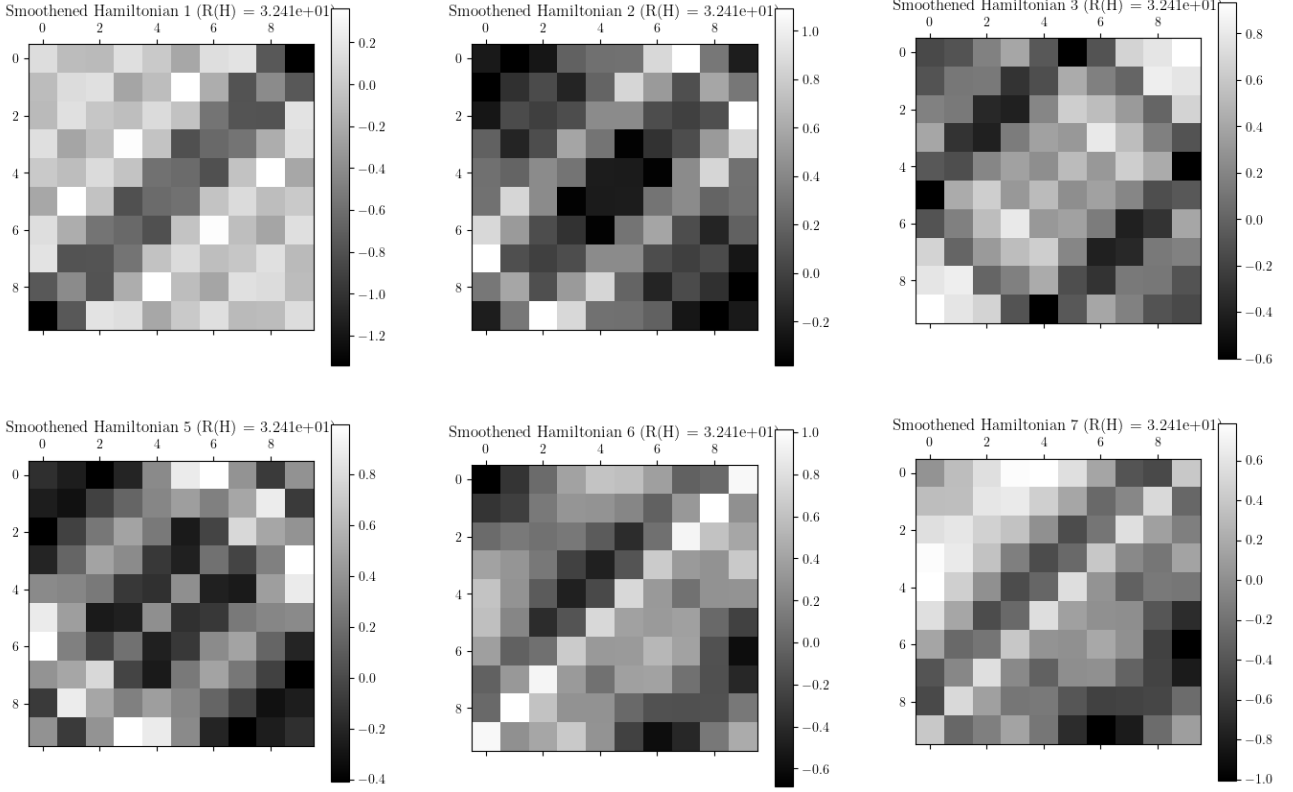


Figure 10: Equally smooth(est) representations of an absolutely degenerate Hamiltonian.

11 Properties of unique smoothest representation

We will now explore some properties of the smoothest representation of a Hamiltonian. We will first explore the properties of the smoothest representation when the orthogonal transformation E_A diagonalizes the Hamiltonian H and H^2 simultaneously. This includes the case when H is not absolutely degenerate.

If E_A diagonalizes H , then the smoothest representation of the Hamiltonian $S(H)$ is given by:

$$S(H) = E_B^T D_H E_B \quad (41)$$

where D_H is the diagonal matrix with eigenvalues of H arranged in decreasing order of their absolute values. For the choice of R as the forward difference matrix, the matrix $B = R^T R$ is the discrete 1D Laplacian with Neumann boundary conditions. Now the eigenvalues and eigenvectors of the matrix B are given by:

$$\lambda_k^B = 2 - 2 \cos\left(\frac{k\pi}{n}\right) \text{ for } 0 \leq k \leq n-1 \quad (42)$$

$$v_0 = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \quad v_k(j) = \sqrt{\frac{2}{n}} \cos\left(\frac{k\pi(j + \frac{1}{2})}{n}\right) \text{ for } 1 \leq k \leq n-1 \text{ and } 0 \leq j \leq n-1 \quad (43)$$

Since B is the discrete Laplacian, $S(H)$ is the **inverse discrete cosine transform** of D_H . Now, the orthogonal matrix E_B is constructed from the eigenvectors of B and hence $S(H)$ can be expressed as:

$$S(H) = E_B^T D_H E_B = \sum_{k=0}^{n-1} \lambda_k^H v_k v_k^T \quad (44)$$

$$\Rightarrow S(H)_{i,j} = \frac{\lambda_0^H}{n} + \frac{2}{n} \sum_{k=1}^{n-1} \lambda_k^H \cos\left(\frac{k\pi(i + \frac{1}{2})}{n}\right) \cos\left(\frac{k\pi(j + \frac{1}{2})}{n}\right) \quad (45)$$

$$= \frac{\lambda_0^H}{n} + \frac{1}{n} \sum_{k=1}^{n-1} \lambda_k^H \left(\cos\left(\frac{k\pi(i+j+1)}{n}\right) + \cos\left(\frac{k\pi(i-j)}{n}\right) \right) \quad (46)$$

This immediately gives us the following symmetries of $S(H)$:

- $S(H)$ is **symmetric**, i.e., $S(H)_{i,j} = S(H)_{j,i}$ for all i, j .
- $S(H)$ is **centrosymmetric**, i.e., $S(H)_{i,j} = S(H)_{n-i-1, n-j-1}$ for all i, j .
- $S(H)$ is **anti-diagonal symmetric**, i.e., $S(H)_{i,j} = S(H)_{n-j-1, n-i-1}$ for all i, j .
- $S(H)$ can be expressed as a sum of a **Hankel** matrix ($\mathcal{H}$) and a **Toeplitz** matrix ($\mathcal{T}$).

$$S(H) = \mathcal{T} + \mathcal{H} - \frac{\lambda_0^H}{n} I$$

$$\mathcal{T}_{i,j} = \frac{1}{n} \sum_{k=0}^{n-1} \lambda_k^H \cos\left(\frac{k\pi(i-j)}{n}\right)$$

$$\mathcal{H}_{i,j} = \frac{1}{n} \sum_{k=0}^{n-1} \lambda_k^H \cos\left(\frac{k\pi(i+j+1)}{n}\right)$$

12 Example: Smoothest representation of a Hamiltonian with equally spaced energy levels

Let us consider a $n \times n$ Hamiltonian with equally spaced eigenvalues, i.e., $\lambda_k^H = E_0 + k\Delta$ for $k = 0, 1, \dots, n-1$ where E_0 is the ground state energy and Δ is the spacing between energy levels.

It can be shown that the smoothest representation of such a Hamiltonian is given by:

$$S(H)_{i,j} = \left(E_0 + \frac{n\Delta}{2} - \Delta\right) \delta_{i,j} + \frac{\Delta}{2n} \begin{cases} \operatorname{cosec}^2\left(\frac{\pi(i-j)}{2n}\right) & \text{if } i-j \text{ is odd} \\ \operatorname{cosec}^2\left(\frac{\pi(i-j+1)}{2n}\right) & \text{if } i-j \text{ is even} \end{cases} \quad (47)$$

If we plot the variation of diagonal entries, Toeplitz entries and Hankel entries corresponding to the smoothest representation of such a Hamiltonian, we get the following plots:

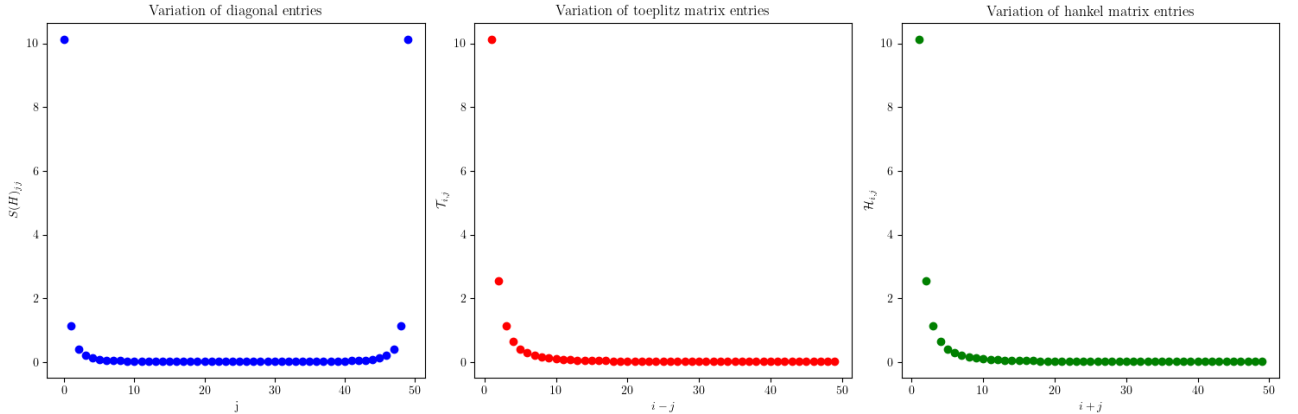


Figure 11: Variation of diagonal entries, Toeplitz entries and Hankel entries corresponding to the smoothest representation.

The net effect of all these properties is the smoothest representation of the Hamiltonian which is concentrated around the main diagonal, as shown in the figure 12.

In addition, the smoothest representation has certain structural properties in some cases :

- If n is odd, then the anti-diagonal of $S(H)$ is a constant. And,

$$\Delta = \text{Anti-diagonal} \times 2n$$

$$E_0 = \text{Anti-diagonal} \times (2-n)n$$

So, we can recover complete information about the Hamiltonian from the anti-diagonal of its smoothest representation.

- If n is even and $\frac{n}{2}$ is odd, if we inscribe a square of dimensions $n \times n$ in the smoothest representation titled through an angle of $\frac{\pi}{4}$ about the center, then the all entries along the square is a constant. And, the constant times the dimension of the square gives us the spacing between the energy levels and the constant times $\frac{n(2-n)}{2}$ gives us the ground state energy. Hence, we can recover complete information about the Hamiltonian from the diamond inscribed in the smoothest representation.

- If n is even and $\frac{n}{2}$ is even, we don't see any such structure in the smoothest representation. However, we can still recover the ground state energy and the spacing between the energy levels as follows:

$$\Delta = 2nS(H)_{0,1}\sin^2\left(\frac{\pi}{2n}\right)$$

$$E_0 = S(H)_{0,0} - S(H)_{0,1} + \left(\frac{2-n}{2}\right)\Delta$$

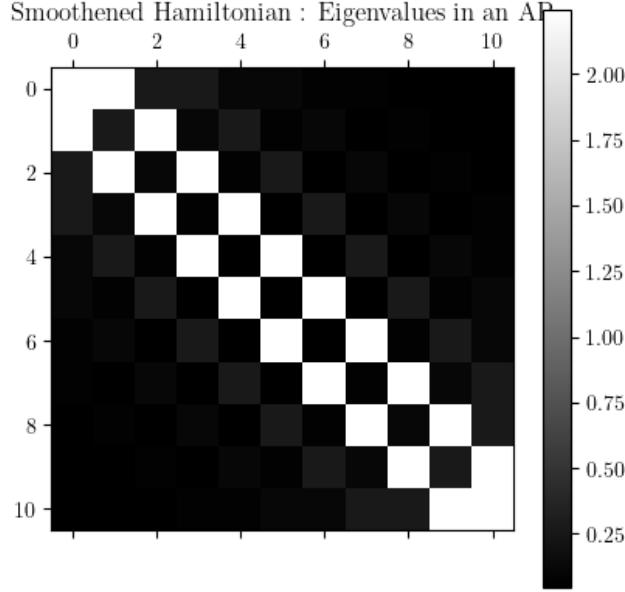


Figure 12: Matrix plot of $S(H)$

13 Image-like representation of Hamiltonians and Discrete Cosine Transform

We wanted to find an orthogonal transformation that would make the Hamiltonian “smooth” and “image-like”. In previous sections, we proposed a method to arrive at the smoothest form of a Hamiltonian but is that representation really image-like? That is, does it look like a natural image?

In the previous section, we saw that the smoothest form of a Hamiltonian (or one of its smoothest forms if it is absolutely degenerate) is the inverse discrete cosine transform of the diagonalized Hamiltonian. The discrete cosine transform (DCT) is a widely used transformation in image processing, particularly in image compression. It maps a signal ($n \times n$ image) to a $n \times n$ matrix of DCT coefficients. One of the salient features of DCT of natural images is energy compaction, which means that most of the relevant visual information is concentrated in a few low frequency coefficients[3]. Also, the human eye is more sensitive to the low order of frequency signals than the high order of frequency signals[2]. So one can compress the image by discarding the high frequency coefficients and then reconstructing the image using the inverse DCT. This is the basis of JPEG compression. We have seen that the DCT of the smoothest form of a Hamiltonian is the diagonalized Hamiltonian with eigenvalues arranged in decreasing order of their absolute values. This means that the image has the property of energy compaction, i.e., most of the relevant information is concentrated in a few low frequency coefficients like natural images.

14 Reduced Smooth Representations of Hamiltonians

In section(11), we saw that the smoothest representation (or one of the smoothest representations) of a Hamiltonian is the sum of a Hankel matrix and a Toeplitz matrix. This means that for an $n \times n$ Hamiltonian, we require only $2n$ parameters to represent the smoothest form of the Hamiltonian and the smoothest form has a lot of symmetries. This motivates us to ask the question - *Can we define a reduced smooth representation of a given Hamiltonian that captures the spectrum of the Hamiltonian?*

Given a $n \times n$ Hamiltonian H , we want to define a $m \times m$ smooth matrix $S_r(H)$ such that $S_r(H)$ captures the spectrum of H and m is as small as possible. Since H is a symmetric (or Hermitian) matrix, its spectrum is a set of n real numbers. We want to define a $m \times m$ matrix $S_r(H)$ such that :

- $S_r(H)$ contains complete information about the spectrum of H .
- $S_r(H)$ is a smooth matrix.
- $m \geq \sqrt{n}$ is as small as possible.

In the previous section, we saw that the discrete cosine transform of a smooth natural image is concentrated in a few low frequency coefficients, i.e., in the top left corner of the DCT matrix. We can use this property to define a reduced smooth representation of a Hamiltonian.

Fix m such that $m \geq \sqrt{n}$. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of H such that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$. We define $S_{dct}(H)$ as the $m \times m$ matrix given by:

$$S_{dct}(H) = \begin{pmatrix} \lambda_1 & \lambda_2 & \lambda_4 & \cdots & 0 & 0 \\ \lambda_3 & \lambda_5 & \cdots & \cdots & 0 & 0 \\ \lambda_6 & \cdots & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \cdots & 0 & 0 \end{pmatrix} \quad (48)$$

That is, the dominant eigenvalues of H are arranged in the top left corner of $S_{dct}^{(m)}(H)$ along anti-diagonals and the rest of the entries are zero. We can see that $S_{dct}^{(m)}(H)$ captures the spectrum of H . Now, let us define $S_r^{(m)}(H)$ as the inverse discrete cosine transform of $S_{dct}^{(m)}(H)$. Since dominant entries are concentrated in the top left corner of $S_{dct}^{(m)}(H)$, the inverse discrete cosine transform will produce a smooth matrix $S_r^{(m)}(H)$ [3]. Hence, $S_r^{(m)}(H)$ is smooth and its discrete cosine transform captures the spectrum of H .

Note that the proposed method of obtaining $S_r^{(m)}(H)$ doesn't impose any constraints on m . So, we can choose m to be the smallest integer greater than $\sqrt{n}$, i.e., $m = \lceil \sqrt{n} \rceil$. One can also choose m to be larger than $\lceil \sqrt{n} \rceil$, this will make the top left corner of $S_{dct}^{(m)}(H)$ more dominant and hence the smoothness of $S_r^{(m)}(H)$ will increase.

The adjacent figure(13) shows how the roughness of the reduced smooth representation $S_r^{(m)}(H)$ decreases as m increases for a random 100×100 Hamiltonian. A log-log plot of the roughness of $S_r^{(m)}(H)$ against m showed linear dependence, indicating power law scaling of the roughness with respect to m . The fitted line and fitting parameters are shown in the figure(13).

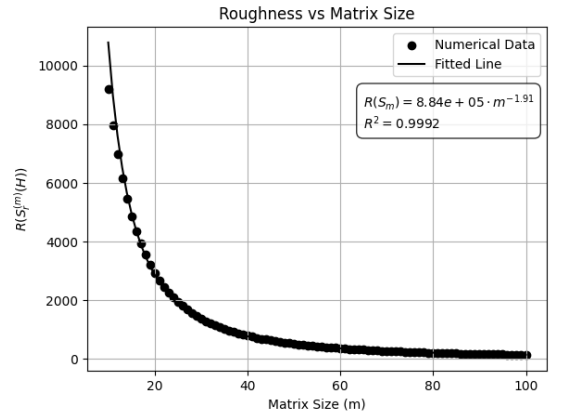


Figure 13: Example of a reduced smooth representation $S_r^{(m)}(H)$ for a Hamiltonian H .

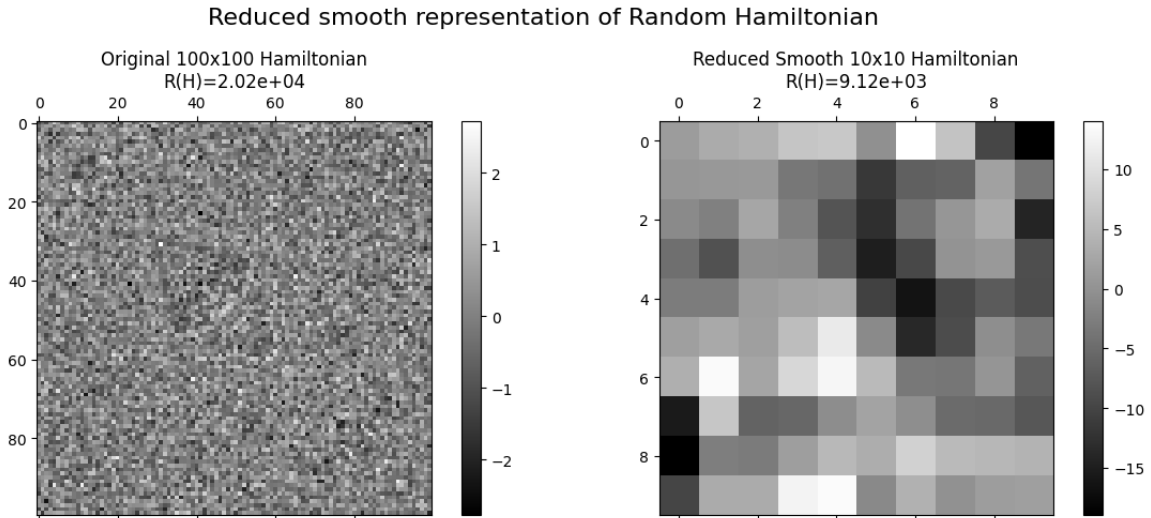


Figure 14: Reduced smooth representation of a random 100×100 Hamiltonian.

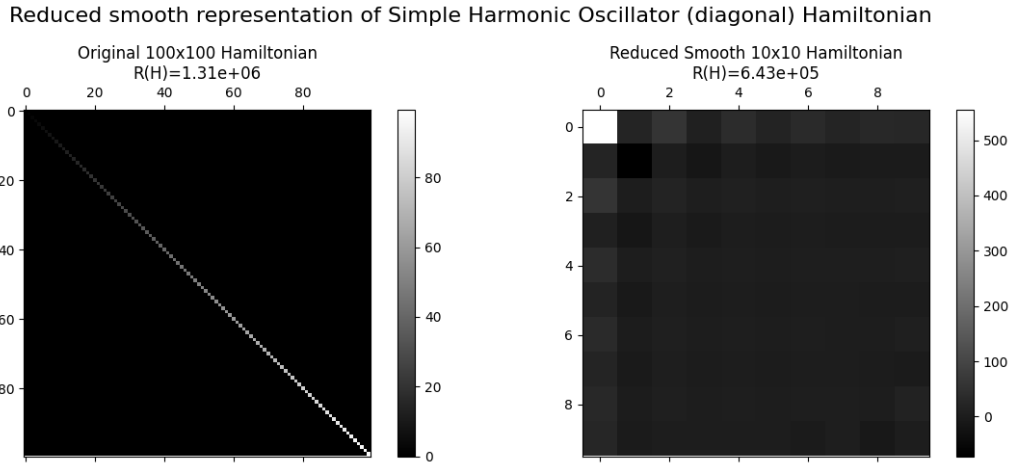


Figure 15: Reduced smooth representation of a 100×100 Simple Harmonic Oscillator Hamiltonian

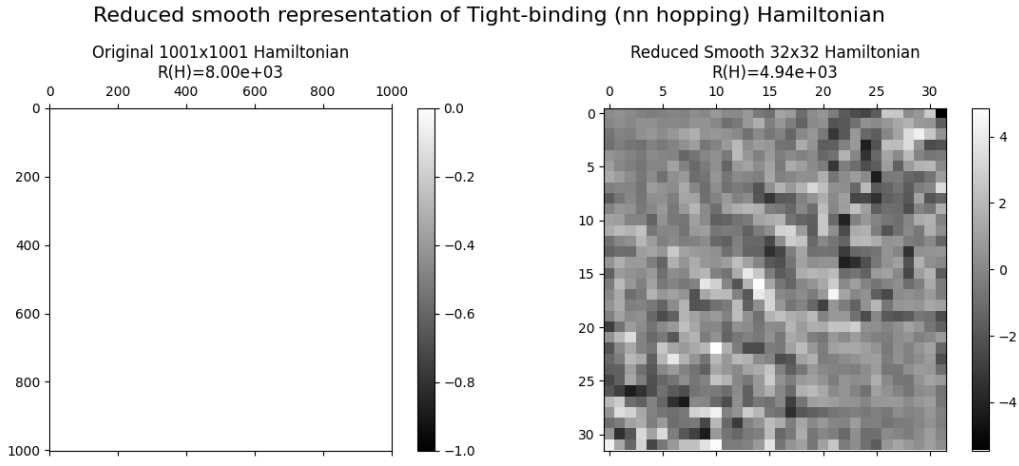


Figure 16: Reduced smooth representation of a 1001×1001 Tight Binding Hamiltonian with nearest neighbor interactions

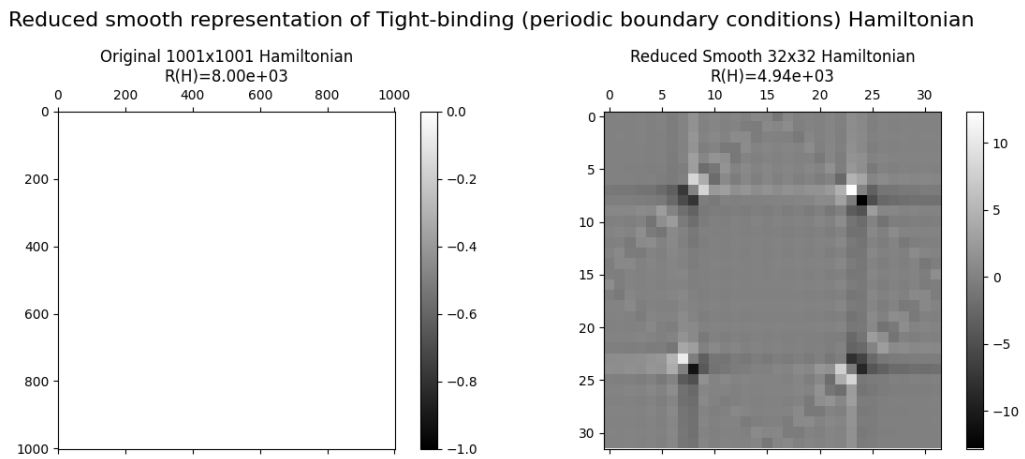


Figure 17: Reduced smooth representation of a 1001×1001 Tight Binding Hamiltonian with nearest neighbor interactions and periodic boundary conditions

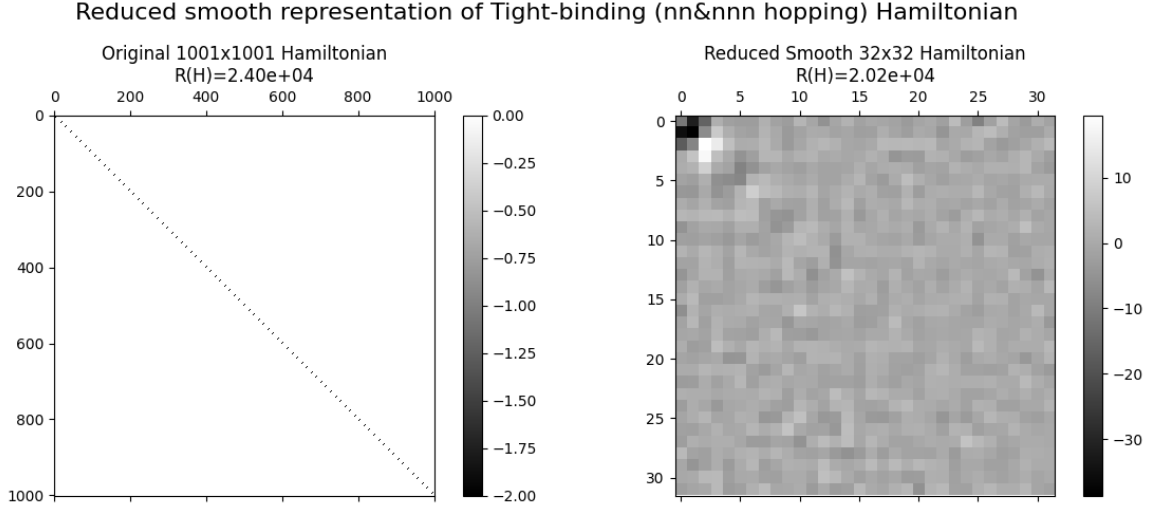


Figure 18: Reduced smooth representation of a 1001×1001 Tight Binding Hamiltonian with nearest and next nearest neighbor interactions

In the above examples, we have chosen $m = \lceil \sqrt{n} \rceil$ for each Hamiltonian. On increasing m , we find that the image becomes smoother, but the overall structure of the image remains the same. For example, the reduced smooth representation of the Tight Binding Hamiltonian with nearest neighbor interactions and periodic boundary conditions is shown in figure(19) for $m = 100$. We can see that the image is smoother than the one shown in figure(17), but the overall structure of the image remains the same.

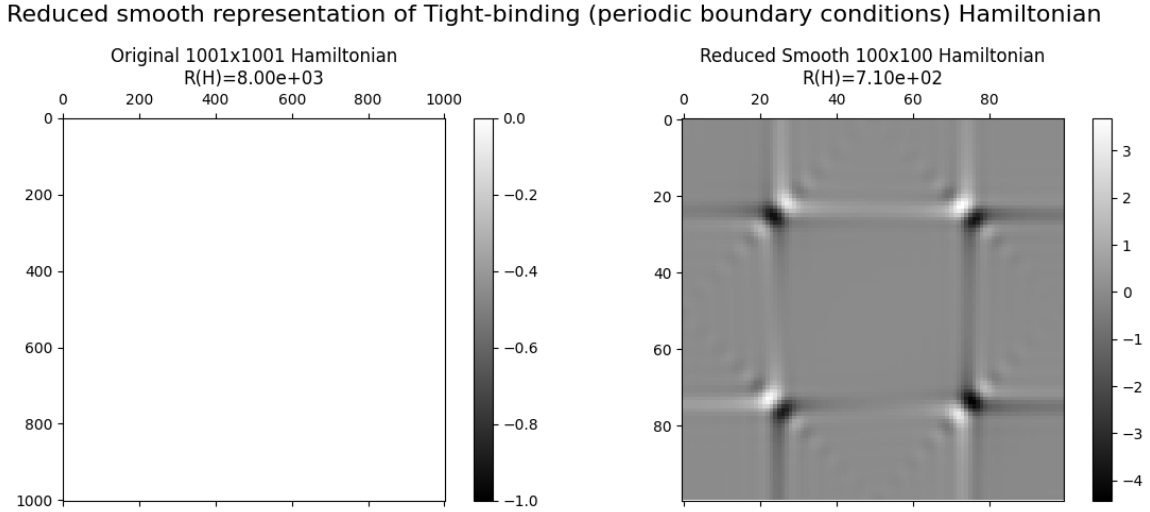


Figure 19: Reduced smooth representation of a 1001×1001 Tight Binding Hamiltonian with nearest neighbor interactions and periodic boundary conditions with $m = 100$.

15 Smoothing Asymmetric Matrices

When the Hamiltonian is not symmetric, the definition of roughness is given by eq(3). If one wants to find an orthogonal transformation that minimizes the roughness of an asymmetric Hamiltonian, the Lagrangian in eq(17) can be modified as follows:

$$\mathcal{L}(U, \Lambda) = Tr[U^T H H^T U B] + Tr[U^T H^T H U B] + Tr[\Lambda(U^T U - I)] \quad (49)$$

If we minimize the above Lagrangian with respect to U and Λ , we finally arrive at the following closed form solution for the smoothest form of an asymmetric Hamiltonian:

$$S(H) = E_B^T E_A H E_A^T E_B \quad (50)$$

where E_A and E_B are the orthogonal matrices constructed from the eigenvectors of $A = H^T H + H H^T$ and $B = R^T R$ respectively. The optimal ordering of the eigenvectors of A and B is such that the eigenvalues of A

increases and eigenvalues of B decreases, as we go down the diagonal in D_A and D_B respectively.

16 Blurring Images of Hamiltonians

In image processing, blurring is a technique used to reduce noise and detail in an image. This is typically achieved by replacing each pixel's value with a weighted average of its neighboring pixels' values. If one does the same to a noisy Hamiltonian in some arbitrary basis, it might reduce the noise, but it can completely distort the eigenvalues and eigenvectors of the Hamiltonian.

16.1 1 dimensional blurring

Now that we have a smooth image representation of a Hamiltonian, one can ask how blurring the image affects the eigenvalues and eigenvectors of the Hamiltonian. If they are not distorted much, then blurring can be used to reduce noise in the Hamiltonian. In this context, we define blurring as follows:

Definition: Let A be a $n \times n$ matrix. We define the blurred version of A of order m as the matrix $A_b^{(m)}$ given by:

$$(A_b^{(m)})_{i,j} = \frac{1}{4m+1} \left(A_{i,j} + \sum_{k=1}^m (A_{i-k,j} + A_{i+k,j} + A_{i,j-k} + A_{i,j+k}) \right) \quad (51)$$

where, the summation is done assuming Neumann (reflecting) boundary conditions (The choice of this boundary condition will become clear later).

To answer the question of how blurring affects the eigenvalues and eigenvectors of a Hamiltonian, we state the following theorems:

Theorem 16.1. *Let H be a $n \times n$ Hamiltonian and $S(H)$ be its smoothest representation. Then the eigenvectors of $S(H)$ remain unchanged under blurring of any order m .*

Proof. Let the eigenvalues of H be $\lambda_0, \lambda_1, \dots, \lambda_{n-1}$ such that $|\lambda_0| \geq |\lambda_1| \geq \dots \geq |\lambda_{n-1}|$. Let $S_b^{(m)}(H)$ be the blurred version of $S(H)$ of order m . We can write the entries of $\tilde{S} = S_b^{(m)}(H)$ as:

$$\tilde{S}_{i,j} = \frac{1}{4m+1} \left(S(H)_{i,j} + \sum_{k'=1}^m (S(H)_{i-k',j} + S(H)_{i+k',j} + S(H)_{i,j-k'} + S(H)_{i,j+k'}) \right) \quad (52)$$

$$= \frac{\lambda_0}{n} + \frac{2}{n(4m+1)} \sum_{k=1}^{n-1} \lambda_k \left(1 + 4 \sum_{k'=1}^m \cos \left(\frac{kk'\pi}{n} \right) \right) \cos \left(\frac{k(i + \frac{1}{2})\pi}{n} \right) \cos \left(\frac{k(j + \frac{1}{2})\pi}{n} \right) \quad (53)$$

where we have used the expression for $S(H)_{i,j}$ derived in section(11). Define:

$$\tilde{\lambda}_k = \frac{1}{4m+1} \left(1 + 4 \sum_{k'=1}^m \cos \left(\frac{kk'\pi}{n} \right) \right) \lambda_k \quad (54)$$

We can see that $\tilde{S}$ can be expressed as:

$$\tilde{S}_{i,j} = \frac{\tilde{\lambda}_0}{n} + \frac{2}{n} \sum_{k=1}^{n-1} \tilde{\lambda}_k \cos \left(\frac{k(i + \frac{1}{2})\pi}{n} \right) \cos \left(\frac{k(j + \frac{1}{2})\pi}{n} \right) \quad (55)$$

This means that $S_b^{(m)}(H)$ is the inverse discrete cosine transform of the diagonal matrix $D = \text{diag}(\tilde{\lambda}_0, \tilde{\lambda}_1, \dots, \tilde{\lambda}_{n-1})$. Hence, it is diagonalized by the same orthogonal matrix E_B that diagonalizes $S(H)$. Hence, the eigenvectors of $S(H)$ remain unchanged under blurring of any order m . $\square$

In the above proof, one requires Neumann boundary conditions to ensure that the blurred matrix is the inverse DCT of a diagonal matrix.

Theorem 16.2. *Let H be a $n \times n$ Hamiltonian and $S(H)$ be its smoothest representation. Then the eigenvalues of $S(H)$ change under blurring of order m as follows:*

$$\tilde{\lambda}_k = \frac{1}{4m+1} \left(2 \sin \left(\frac{k\pi}{2n} (2m+1) \right) \csc \left(\frac{k\pi}{2n} \right) - 1 \right) \lambda_k \quad (56)$$

where $\tilde{\lambda}_k$ are the eigenvalues of the blurred matrix $S_b^{(m)}(H)$.

Proof. From the proof of the previous theorem, we can see that the eigenvalues of the blurred matrix $S_b^{(m)}(H)$ are given by $\tilde{\lambda}_k$ where $\tilde{\lambda}_k$ is defined as:

$$\tilde{\lambda}_k = \frac{1}{4m+1} \left(1 + 4 \sum_{k'=1}^m \cos \left(\frac{kk'\pi}{n} \right) \right) \lambda_k \quad (57)$$

We can simplify the summation using the formula for the sum of cosines:

$$\tilde{\lambda}_k = \frac{1}{4m+1} \left(2 \sin \left(\frac{k\pi}{2n} (2m+1) \right) \csc \left(\frac{k\pi}{2n} \right) - 1 \right) \lambda_k \quad (58)$$

This completes the proof. $\square$

Theorem 16.3. *Let H be a $n \times n$ Hamiltonian and $S(H)$ be its smoothest representation. Then the eigenvalues and eigenvectors of $S(H)$ remain unchanged under blurring of any order m as $n \rightarrow \infty$.*

Proof. From the previous theorem, we can see that the eigenvalues of the blurred matrix $S_b^{(m)}(H)$ are given by:

$$\tilde{\lambda}_k = \frac{1}{4m+1} \left(2 \sin \left(\frac{k\pi}{2n} (2m+1) \right) \csc \left(\frac{k\pi}{2n} \right) - 1 \right) \lambda_k \quad (59)$$

Taking the limit as $n \rightarrow \infty$, we get:

$$\lim_{n \rightarrow \infty} \tilde{\lambda}_k = \frac{1}{4m+1} (2(2m+1) - 1) \lambda_k = \lambda_k \quad (60)$$

This means that the eigenvalues of the blurred matrix $S_b^{(m)}(H)$ converge to the eigenvalues of $S(H)$ as n increases. From the proof of the first theorem, we know that the eigenvectors of $S(H)$ remain unchanged under blurring of any order m . Hence, as n increases, both the eigenvalues and eigenvectors of $S_b^{(m)}(H)$ converge to those of $S(H)$. $\square$

Remark: In the above proof, we have fixed k to be finite and taken the limit as $n \rightarrow \infty$. If we let k increase with n , such that $\lim_{n \rightarrow \infty} \frac{k}{n} = t$ where t is a finite constant, then the eigenvalues of the blurred matrix $S_b^{(m)}(H)$ converge to:

$$\lim_{n \rightarrow \infty} \tilde{\lambda}_k = \frac{2 \frac{\sin(\frac{\pi}{2}(2m+1)t)}{\sin(\frac{\pi}{2}t)} - 1}{4m+1} \lambda_k \quad (61)$$

For finite k , we have $t = 0$ and hence the eigenvalues converge to λ_k . This means that the eigenvalues of the blurred matrix $S_b^{(m)}(H)$ with larger absolute values converge to the eigenvalues of $S(H)$ as n increases, while the eigenvalues with smaller absolute values converge to a fraction of the eigenvalues of $S(H)$.

16.2 2 dimensional blurring: Box blur

In the previous section, we defined blurring along one dimension, that is, we replaced each pixel's value with the average of its neighboring pixels along the same row and column. In this section, we extend this idea to two dimensions by using a box blur.

Let us consider an example of box blurring of order m , that is, we replace each pixel's value with the average of its neighboring pixels in a $(2m+1) \times (2m+1)$ square centered at that pixel.

$$(A_{bb}^{(m)})_{i,j} = \frac{1}{(2m+1)^2} \sum_{k=-m}^m \sum_{l=-m}^m A_{i+k,j+l} \quad (62)$$

where, the summation is done assuming Neumann (reflecting) boundary conditions.

The theorems stated in the previous section can be extended to box blurring as follows:

1. **Eigenvectors:** The eigenvectors of $S(H)$ remain unchanged under box blurring.
2. **Eigenvalues:** The eigenvalues of $S(H)$ change under box blurring of order 1 as follows:

$$\tilde{\lambda}_k = \frac{1}{(2m+1)^2} \left(2 \sin \left(\frac{k\pi}{2n} (2m+1) \right) \csc \left(\frac{k\pi}{2n} \right) \right)^2 \lambda_k \quad (63)$$

where $\tilde{\lambda}_k$ are the eigenvalues of the blurred matrix $S_{bb}(H)$.

3. **Asymptotic behaviour:** For any $k \in \mathbb{N}$, the k^{th} eigenvalue and eigenvector of $S(H)$ remain unchanged under box blurring as $n \rightarrow \infty$.

So we find that even though we extended the blurring to two dimensions, the transformation of the eigenvalues and eigenvectors remains the same as in the one dimensional case, except for a different multiplicative factor in the transformation of the eigenvalues.

The following figure shows how different eigenvalues of a Hamiltonian are scaled under box blurring of different orders m .

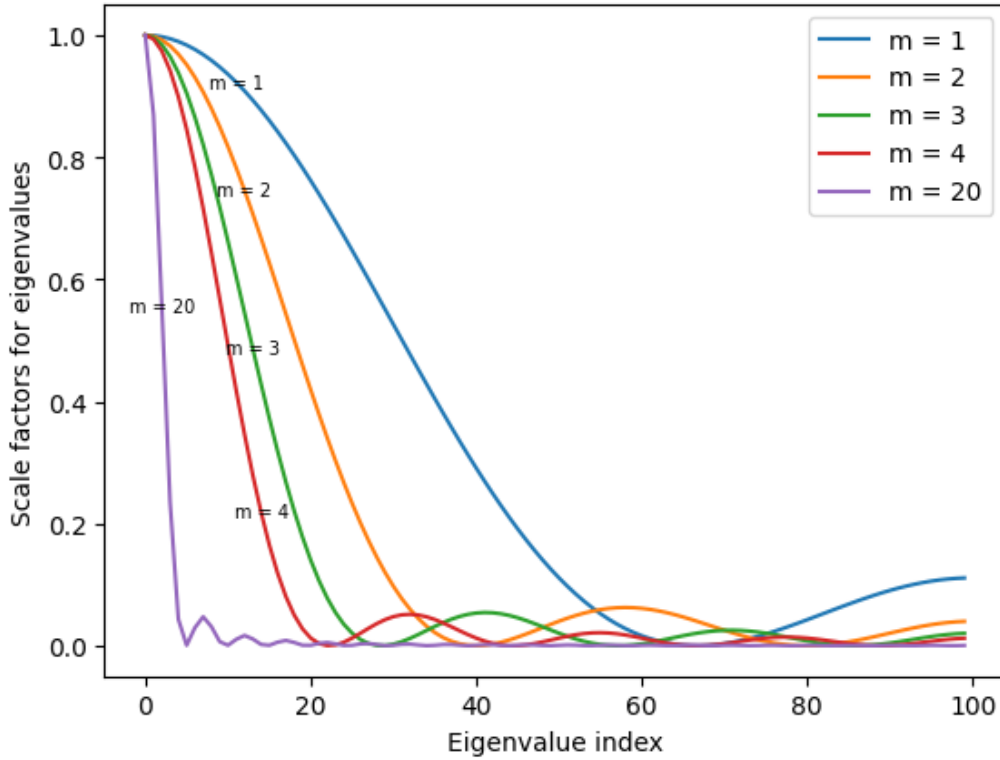


Figure 20: Scale factors for different eigenvalues (ordered in decreasing order of absolute values) under box blurring of different orders m .

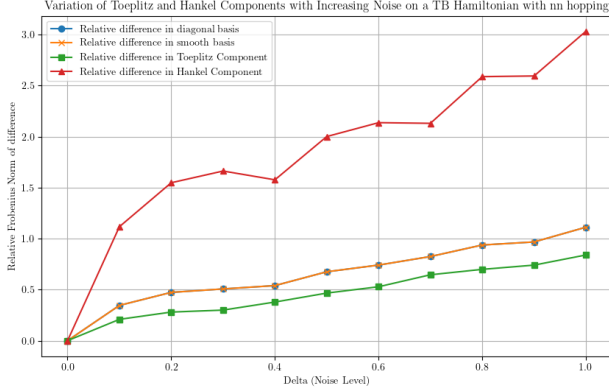
We can see that the eigenvalues with small absolute values (which contribute to high frequency variations in the image) are scaled down more than the eigenvalues with large absolute values (which contribute to low frequency variations in the image). And as m increases, the eigenvalues with small absolute values are scaled down more. This means that box blurring smoothens the image by reducing the high frequency variations. If high frequency noise is present in the image, box blurring reduces the noise by scaling down the eigenvalues that contribute to high frequency variations. But this doesn't correct the eigenvalues, it just scales them down.

17 Smooth Images of Noisy Hamiltonians

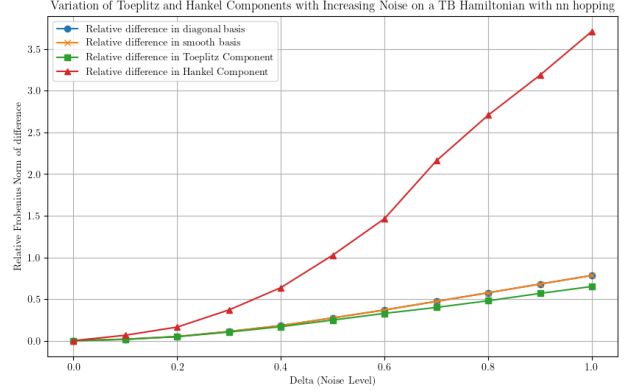
Suppose we have a Hamiltonian H which is corrupted by noise, i.e., $H = H_0 + \delta\epsilon$ where H_0 is the original Hamiltonian, ϵ is the noise matrix and δ is a small parameter. We are interested in exploring whether the smoothest form of the noisy Hamiltonian $S(H)$ can be used to recover the spectrum of the original Hamiltonian H_0 .

To begin with, let us explore if the smoothest form of the noisy Hamiltonian $S(H)$ is closer to $S(H_0)$ than diagonalized Hamiltonian $D(H)$ is to $D(H_0)$. That is, we want to compare $\|S(H) - S(H_0)\|$ with $\|D(H) - D(H_0)\|$. But, we know that for any matrix A , $\|A\| = \sqrt{\text{Tr}[A^T A]}$. Since the smoothest form of a Hamiltonian is a similarity transformation of the diagonalized Hamiltonian, the norm remains unchanged, i.e., $\|S(H) - S(H_0)\| = \|D(H) - D(H_0)\|$. We had seen that the smoothest form of a Hamiltonian has two components - a Hankel matrix and a Toeplitz matrix. Even though $S(H)$ is as distant from $S(H_0)$ as $D(H)$ is from $D(H_0)$, one can ask whether the Hankel and Toeplitz components of $S(H)$ are closer to the Hankel and Toeplitz components of $S(H_0)$. To explore this, we study the difference between the Hankel and Toeplitz components of $S(H)$ and $S(H_0)$ numerically for a few Hamiltonians.

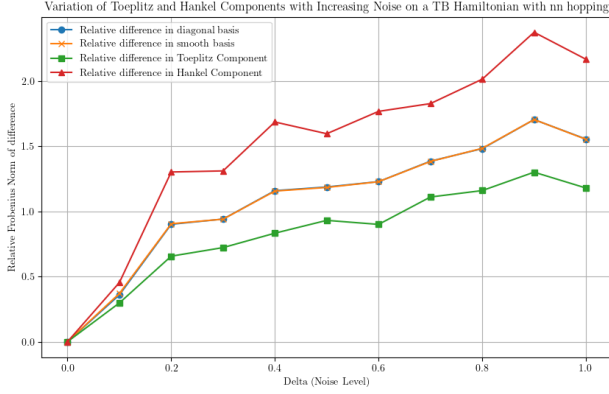
In the following figures, we show the variation of relative difference between the Hankel and Toeplitz components of $S(H)$ and $S(H_0)$, i.e., $\frac{\|\mathcal{H}(H) - \mathcal{H}(H_0)\|}{\|\mathcal{H}(H_0)\|}$ and $\frac{\|\mathcal{T}(H) - \mathcal{T}(H_0)\|}{\|\mathcal{T}(H_0)\|}$, as a function of noise parameter δ for a few Hamiltonians with $\|H_0\| = 1$ and $\|\epsilon\| = 1$.



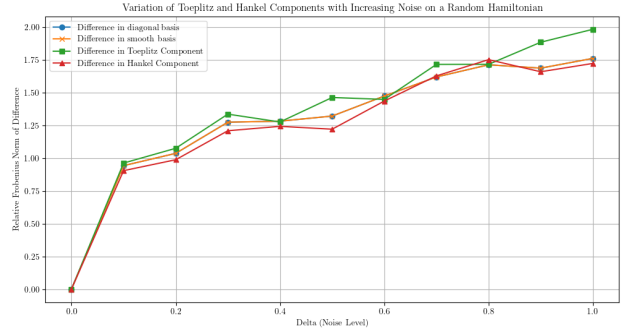
(a) H_0 : Tight Binding Hamiltonian with nearest neighbor interactions



(b) H_0 : Diagonalized Simple Harmonic Oscillator Hamiltonian



(c) H_0 : Heisenberg Hamiltonian



(d) H_0 : Random Hamiltonian

Figure 21: Relative difference between Hankel(red) component, Toeplitz(green) component and the smooth image(orange) for noisy Hamiltonians as a function of noise parameter δ .

We can see that for most of the Hamiltonians, the Toeplitz component is closer to the original Toeplitz component and the Hankel component is distant from the original Hankel component, compared to the relative difference between the smooth image (or the diagonalized form) of H and H_0 . But this trend ceases to hold for the random Hamiltonian as shown in figure(21d).

18 Noisy Transmission of Images of Hamiltonians: Recovery of Spectrum

Consider the problem of transmitting a Hamiltonian H_0 over a noisy channel. The receiver obtains a noisy version of the Hamiltonian, $H = H_0 + \epsilon$, where ϵ is a symmetric noise matrix. If one sends the Hamiltonian in its smooth image representation, the additive noise typically degrades the image's structure, symmetries and smoothness. Since we know different structural properties and symmetries of the image representation of a Hamiltonian, we can use this prior knowledge to obtain a better estimate of the image, and hence the spectrum of the original Hamiltonian.

Let us lay out the problem more formally. Let $\mathbb{S}_n$ be the space of images of all $n \times n$ symmetric matrices.

$$\mathbb{S}_n = \{S \in \mathbb{R}^{n \times n} : S_{i,j} = \frac{\lambda_0}{n} + \frac{2}{n} \sum_{k=1}^{n-1} \lambda_k \cos\left(\frac{k(i + \frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j + \frac{1}{2})\pi}{n}\right), \lambda_k \in \mathbb{R} \forall 1 \leq k \leq n-1\} \quad (64)$$

Note that the space $\mathbb{S}_n$ is a linear subspace of $\mathbb{R}^{n \times n}$ of dimension n . Given a noisy image $I \in \mathbb{R}^{n \times n}$, we want to find the closest image $S \in \mathbb{S}_n$ to I . This can be formulated as the following optimization problem:

$$S = \arg \min_{\tilde{S} \in \mathbb{S}_n} \|I - \tilde{S}\|^2 \quad (65)$$

Finding the closest image S is equivalent to finding $(\lambda_0, \lambda_1, \dots, \lambda_{n-1}) = (n\alpha_0, \frac{n}{2}\alpha_1, \frac{n}{2}\alpha_2, \dots, \frac{n}{2}\alpha_{n-1}) \in \mathbb{R}^n$ such that:

$$S_{i,j} = \frac{\lambda_0}{n} + \frac{2}{n} \sum_{k=1}^{n-1} \lambda_k \cos\left(\frac{k(i+\frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j+\frac{1}{2})\pi}{n}\right) = \sum_{k=0}^{n-1} \alpha_k \cos\left(\frac{k(i+\frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j+\frac{1}{2})\pi}{n}\right) \quad (66)$$

We want to minimize the following objective function:

$$\mathcal{F}(S) = \|I - S\|^2 = \text{Tr}[(S - I)^T(S - I)] = \text{Tr}[I^2 + S^2 - 2SI] \quad (67)$$

$$\mathcal{F}(\alpha_0, \alpha_1, \dots, \alpha_{n-1}) = \text{Tr}[I^2] + \sum_{k=0}^{n-1} \frac{n^2}{(2 - \delta_{k0})^2} \alpha_k^2 - 2 \sum_{i,j,k=0}^{n-1} I_{ij} \cos\left(\frac{k(i+\frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j+\frac{1}{2})\pi}{n}\right) \alpha_k \quad (68)$$

To minimize the above objective function, we set the gradient to zero and solve for α_k :

$$\frac{\partial \mathcal{F}}{\partial \alpha_k} = \frac{2n^2}{(2 - \delta_{k0})^2} \alpha_k - 2 \sum_{i,j=0}^{n-1} I_{ij} \cos\left(\frac{k(i+\frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j+\frac{1}{2})\pi}{n}\right) = 0 \quad (69)$$

$$\Rightarrow \alpha_k = \frac{(2 - \delta_{k0})^2}{n^2} \sum_{i,j=0}^{n-1} I_{ij} \cos\left(\frac{k(i+\frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j+\frac{1}{2})\pi}{n}\right) \quad (70)$$

$$\Rightarrow \lambda_k = \frac{(2 - \delta_{k0})}{n} \sum_{i,j=0}^{n-1} I_{ij} \cos\left(\frac{k(i+\frac{1}{2})\pi}{n}\right) \cos\left(\frac{k(j+\frac{1}{2})\pi}{n}\right) \quad (71)$$

Hence, we can find the closest image S to a noisy image I by computing the above λ_k values and constructing S using these values. Therefore, using the smooth image representation of a Hamiltonian for transmission over a noisy channel is beneficial, as we can exploit the structure of the image to recover a better estimate of the original image and hence its spectrum.

In the adjacent figure(22), we show an example of transmitting a random Hamiltonian over a noisy channel. We can see that by projecting the received noisy image onto the space of images of Hamiltonians, we can recover a spectrum that is much closer to the original spectrum than the received noisy spectrum.

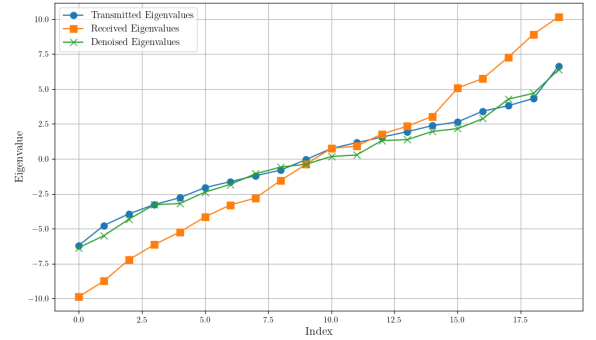


Figure 22: Transmitted, received and recovered spectrum of a random Hamiltonian transmitted over a noisy channel which adds a GOE noise with standard deviation 0.5.

19 Denoising Density Matrices

The denoising scheme proposed in the previous section is not limited to just Hamiltonians, but can be extended to any Hermitian matrix, including density matrices. A central problem in quantum state tomography is to reconstruct the density matrix of a quantum state from a set of measurements. The measurements are often noisy, and the resulting density matrix is not a valid density matrix, i.e., it is not positive semi-definite and has trace not equal to one. Our denoising scheme can be used to reconstruct a valid density matrix from a noisy experimental density matrix.

19.1 Reconstruction of full quantum state

This method is limited to denoising the eigenvalues of the density matrix, and does not correct the eigenvectors of the density matrix. But, a complete reconstruction of the density matrix requires both the eigenvalues and the eigenvectors. What we can do is to use the denoised eigenvalues to construct a new density matrix with the same eigenvectors as the noisy density matrix. The algorithm to do this is as follows:

1. Given a noisy density matrix ρ , diagonalize it as $\rho = UDU^\dagger$ where D is the diagonal matrix of eigenvalues and U is the unitary matrix of eigenvectors.
2. Using the denoising scheme described in the previous section, obtain the denoised eigenvalues $D_{denoised}$.
3. Construct the denoised density matrix as $\rho_{denoised} = UD_{denoised}U^\dagger$.

19.2 Error Analysis

Let $\rho = \rho_0 + \epsilon$ be the noisy density matrix and $\rho_{denoised}$ be the denoised density matrix obtained using the above algorithm. Let U_0 and D_0 be the unitary matrix of eigenvectors and diagonal matrix of eigenvalues of ρ_0 respectively, i.e., $\rho_0 = U_0 D_0 U_0^\dagger$. Let δ denote the difference of the denoised spectrum from the original spectrum, i.e., $\delta = D_{denoised} - D_0$. The error in the denoised density matrix can be quantified as follows:

$$\|\rho_{denoised} - \rho_0\| = \|U D_{denoised} U^\dagger - U_0 D_0 U_0^\dagger\| \quad (72)$$

$$= \|U(D_0 + \delta)U^\dagger - U_0 D_0 U_0^\dagger\| = \|U D_0 U^\dagger + U \delta U^\dagger - U_0 D_0 U_0^\dagger\| \quad (73)$$

$$\leq \|U D_0 U^\dagger - U_0 D_0 U_0^\dagger\| + \|U \delta U^\dagger\| = \|U D_0 U^\dagger - U_0 D_0 U_0^\dagger\| + \|\delta\| \quad (74)$$

The first term $\|U D_0 U^\dagger - U_0 D_0 U_0^\dagger\|$ quantifies the error in the eigenvectors of the denoised density matrix, which is not corrected by the denoising scheme. The second term $\|\delta\|$ quantifies the error in the eigenvalues of the denoised density matrix, which is corrected by the denoising scheme. Hence, we have broken down the error in the denoised density matrix into two components - one due to the eigenvectors and the other due to the eigenvalues and we have shown that a part of the error $\|\delta\|$ is corrected by the denoising scheme and the rest remains as a lower bound on the error in the denoised density matrix.

19.3 Applications of the Denoised Spectrum

Even though the denoised spectrum is not a complete reconstruction of the density matrix, it can be used to obtain some useful information about the quantum state. A few applications of the denoised spectrum are:

- **Purity of the state:** The purity of a quantum state is defined as $Tr[\rho^2]$, which is the sum of the squares of the eigenvalues of the density matrix. The denoised spectrum can be used to compute a better estimate of the purity of the state.
- **Von Neumann Entropy:** The von Neumann entropy of a quantum state is defined as $S(\rho) = -\sum_i \lambda_i \log(\lambda_i)$, where λ_i are the eigenvalues of the density matrix. One can use the denoised spectrum to compute a better estimate of the von Neumann entropy of the state.
- **Probabilistic Envelope of Observables:** Let $\mathcal{O}$ be an observable with eigenvalues $\mu_1 \leq \mu_2 \leq \dots \leq \mu_n$ and ρ be the density matrix of the quantum state which has eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$. This information can be used to compute lower and upper bounds on the expectation value of the observable and hence the probabilistic envelope of the observable. Using von Neumann's trace inequality,

$$\sum_{i=1}^n \lambda_i \mu_{n-i+1} \leq Tr[\rho \mathcal{O}] = \langle \mathcal{O} \rangle \leq \sum_{i=1}^n \lambda_i \mu_i \quad (75)$$

The probabilistic envelope of any observable can be computed using the denoised spectrum of the density matrix and the eigenvalues of the observable.

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